BCSE 0101: Digital Image Processing

Assignment – I

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MM: 100 Submission Due Date: Sept 2, 2022

Note:

Take a printout of this assignment and write your answers in the space provided. Scan and upload the filled answer sheet.

I. Complete the following statements. **[1 x 10 = 10]**

1. The discretization of image data in spatial coordinates is known as Sampling.

2. The number of bits required to store a 1024 x 512 image with 512 gray shades is 4718592 bits. (1024 × 512 × 9 bits)

3. The smallest discernible change in the gray level of an image is called its Gray Level Resolution.

4. When the number of pixels in an image is reduced keeping the number of gray levels in the image constant, we observe checkerboard effect.

5. When the no. of gray-levels in the image is low, the foreground details of the image merge with the background details of the image, causing ridge like structures. This degradation phenomenon is known as false contouring.

6. Mark the m-adjacent path from p to q in the following image.

p

1	0	0	0
1	1	0	0
1	1	1	1
1	0	1	1

q

7. The distance between pixels p & q in the above image:

a. Euclidean distance: $\sqrt{(4-1)^2 + (4-1)^2} = \sqrt{18}$

b. City block distance: $|4-1| + |4-1| = 6$

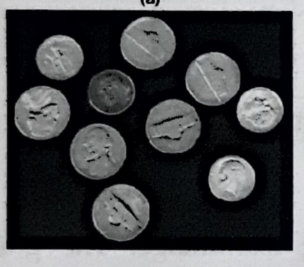
c. Chess board distance: $\max(|4-1|, |4-1|) = 3$

8. Two image subsets S1 & S2 are adjacent if some pixel in S_1 is adjacent to some pixel in S_2.

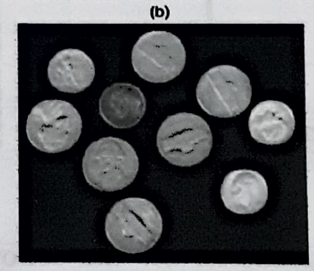
9. Dark characteristics in an image are better enhanced using Power-law transformation(s).

10. Consider the following images. (a) is the original image. On (b) and (c) average filters of different sizes has been applied. On (b) filter of size 3 x 3 has been applied while on (C) filter of size 9 x 9 has been applied.

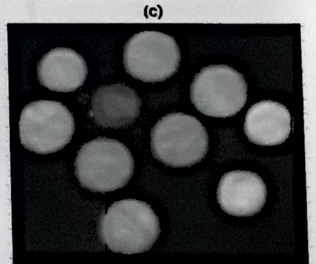
(a)



(b)



(c)



II. Consider two image subsets S1 & S2 as shown in the following figure. For V= {0} determine whether the regions are: i) 4- Adjacent ii) 8-Adjacent iii) m-Adjacent . Give reasons for your answer. [2 x 3 = 6]

S_1				S_2			
1	1	1	1	0	1	0	0
1	1	0	1	0	0	1	1
1	1	0	1	0 q	0	1	1
1	0	0	0 p	1	1	1	1

i) 4- Adjacent – Yes / No ✓ because let p and q be pixels as shown in figure, q is not in the set $N_4(p)$. So they are not 4 adjacent.

ii) 8-Adjacent – Yes / No

Because p and q are 8 adjacent as q is in set $N_8(p)$.

iii) m-Adjacent– Yes / No

Because q is in $N_D(p)$ and the set $N_4(p) \cap N_4(q)$ is empty.

III. Given the following 3 x 3 image, find its bit planes.
Note: There are only 8 Intensity values in the image. [3 x 3 = 9]

001	010	011
100	101	000
111	110	010

1	2	3
4	5	0
7	6	2

1	0	1	0	1	1	0	0	0
0	1	0	0	0	0	1	1	0
1	0	0	1	1	1	1	1	0
Bit Plane 0			Bit Plane 1			Bit Plane 2		

IV. Perform histogram equalization on the following 8x8 image. The gray level distribution of the image is given below. [15]

Gray levels (r_k)	0	1	2	3	4	5	6	7
Number of pixels (p_k)	8	10	10	2	12	16	4	2

i/p Gray Level (r_k)	No. of pixels (n_k)	$p(r_k) = n_k / MN$	Σ	(L-1) Σ (S_k) (L-1)=7	o/p Gray Level	No. of pixels in o/p image
0	8	0.125	0.125	0.875	1	8
1	10	0.15625	0.28125	1.96875	2	10
2	10	0.15625	0.4375	3.0625	3	12
3	2	0.03125	0.46875	3.28125	3	
4	12	0.1875	0.65625	4.59375	5	12
5	16	0.25	0.90625	6.34375	6	16
6	4	0.0625	0.96875	6.78125	7	6
7	2	0.03125	1.0	7.0	7	

V. For a 8 x 8 image as shown below, generate the linear contrast stretched image with minimum gray level 0 and maximum gray level 7. [10]

Note:

$$\frac{r - r_{min}}{r_{max} - r_{min}} = \frac{s - s_{min}}{s_{max} - s_{min}}$$

3	3	3	3	3	3	3	3
3	4	4	4	4	4	4	3
3	4	2	2	2	2	4	3
3	4	2	5	5	2	4	3
3	4	2	5	5	2	4	3
3	4	2	2	2	2	4	3
3	4	4	4	4	4	4	3
3	3	3	3	3	3	3	3

Space for Calculation

$r_{min} = 2$, $r_{max} = 5$, $s_{min} = 0$, $s_{max} = 7$

$$\frac{r-2}{5-2} = \frac{s-0}{7-0} \Rightarrow \frac{r-2}{3} = \frac{s}{7}$$

$$\boxed{s = \frac{7(r-2)}{3}}$$

i/p Gray Level (r_k)	No. of pixels (n_k)	$p(r_k) = n_k / MN$	Σ	$(L-1)\Sigma$ (S_k) $(L-1)=7$	o/p Gray Level	No. of pixels in o/p image
0	8	0.125	0.125	0.875	1	8
1	10	0.15625	0.28125	1.96875	2	10
2	10	0.15625	0.4375	3.0625	3	12
3	2	0.03125	0.46875	3.28125	3	
4	12	0.1875	0.65625	4.59375	5	12
5	16	0.25	0.90625	6.34375	6	16
6	4	0.0625	0.96875	6.78125	7	6
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3	3	3	3	3	3	3	3
3	4	4	4	4	4	4	3
3	4	2	2	2	2	4	3
3	4	2	5	5	2	4	3
3	4	2	5	5	2	4	3
3	4	2	2	2	2	4	3
3	4	4	4	4	4	4	3
3	3	3	3	3	3	3	3

Space for Calculation

$r_{min} = 2$, $r_{max} = 5$, $s_{min} = 0$, $s_{max} = 7$

$$\frac{r-2}{5-2} = \frac{s-0}{7-0} \Rightarrow \frac{r-2}{3} = \frac{s}{7}$$

$$\boxed{s = \frac{7(r-2)}{3}}$$

$r = 0 \Rightarrow s = \frac{7}{3}(-2) \Rightarrow 0$

$r = 1 \Rightarrow s = \frac{7}{3}(-1) \Rightarrow 0$

$r = 2 \Rightarrow s = \frac{7}{3}(0) = 0$

$r = 3 \Rightarrow s = \frac{7}{3} = 2$

$r = 4 \Rightarrow s = \frac{7}{3} \times 2 = 5$

$r = 5 \Rightarrow s = \frac{7}{3} \times 3 = 7$

$r = 6 \Rightarrow s = \frac{7}{3} \times 4 = 7$

$r = 7 \Rightarrow s = 7$

rk	sk
0	0
1	0
2	0
3	2
4	5
5	7
6	7
7	7

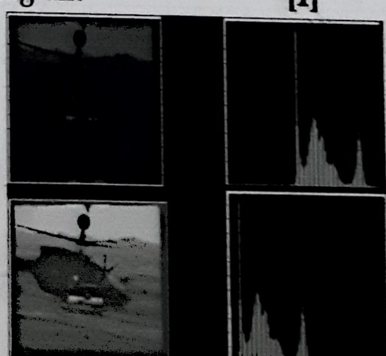
2	2	2	2	2	2	2	2
2	5	5	5	5	5	5	2
2	5	0	0	0	0	5	2
2	5	0	7	7	0	5	2
2	5	0	7	7	0	5	2
2	5	0	0	0	0	5	2
2	5	5	5	5	5	5	2
2	2	2	2	2	2	2	2

Output Image

VI. Match the image to its histogram [1]

a

b



1

2

b → 1

a → 2

VII. Perform Histogram Specification between the following histograms [15]

Original Image Histogram Equalization

I/p Gray Level (r_k)	no. of pixels (n_k)	$p(r_k) = n_k/MN$	Σ	$(L-1)\Sigma$	O/p Gray Level (s)
0	2	0.03125	0.03125	0.21875	0
1	3	0.046875	0.078125	0.546875	1
2	5	0.078125	0.15625	1.09375	1
3	6	0.09375	0.25	1.75	2
4	9	0.140625	0.390625	2.734375	3
5	12	0.1875	0.578125	4.046875	4
6	14	0.21875	0.796875	5.578125	6
7	13	0.203125	1.0	7.0	7

Specified Image Histogram Equalization

I/p Gray Level (r_k)	no. of pixels (n_k)	$p(r_k) = n_k/MN$	Σ	$(L-1)\Sigma$	O/p Gray Level (s)
0	13	0.203125	0.203125	1.421875	1
1	12	0.1875	0.390625	2.734375	3
2	14	0.21875	0.609375	4.265625	4

3	14	0.21875	0.828125	5.796875	6
4	11	0.171875	1.0	7.0	7
5	0	0	1.0	7.0	7
6	0	0	1.0	7.0	7
7	0	0	1.0	7.0	7

r_k	z_q	No. of pixels
0	0	2
1	0	3
2	0	5 = 16
3	0	6
4	1	9
5	2	12
6	3	14
7	4	13

VIII. An 8-bit digital image has a histogram where the gray levels are equally distributed in the range from 160 to 220 (uniform distribution). Sketch the new histograms for each operation as well as the transformation functions for each, and describe the produced effect on the image contrast and brightness.

a) Calculation of the image negative [4 + 4 + 2]

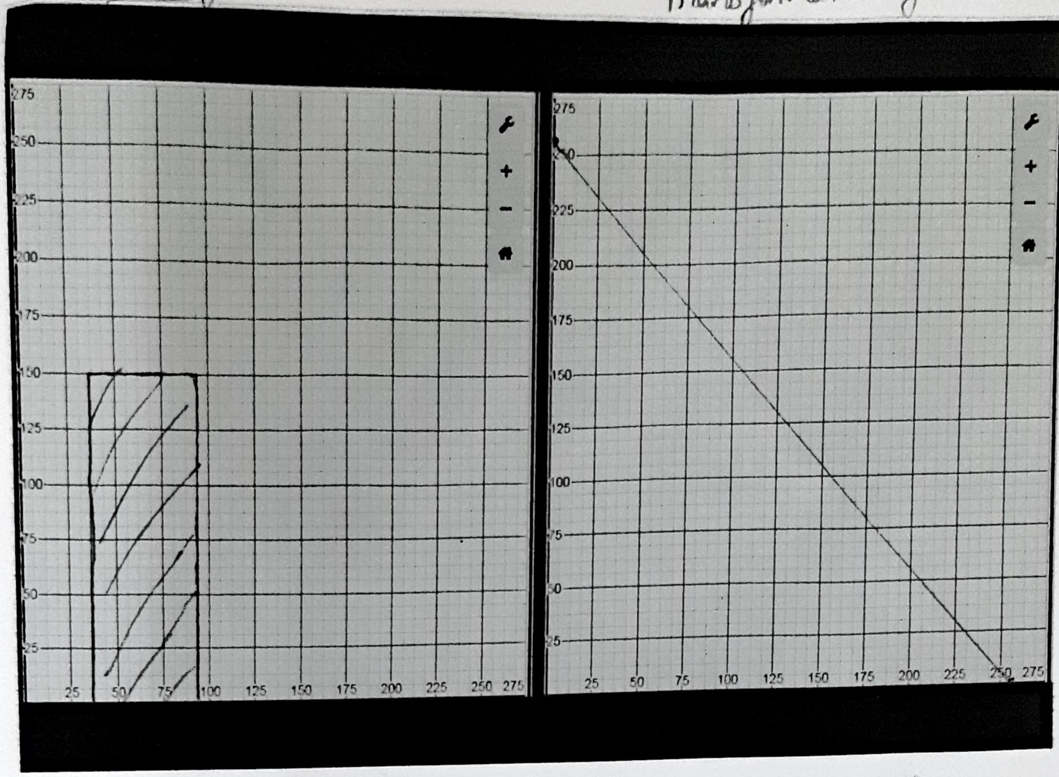
b) Addition of 50 to all pixel gray levels [4 + 4 + 2]

c) Application of a thresholding function where the threshold is selected as gray level 128. [4 + 4 + 2]

(a) The new image will have the order of the gray levels reversed. All the pixels which were 160 will become 255-160 = 95, and all the pixels which were 220 will be 35. The image will be the inverse (the brighter areas will be darker and vice versa). The image will still have low contrast but now it will be darker. The new histogram will have values from 35 to 95. The transformation function is a line with negative slope.

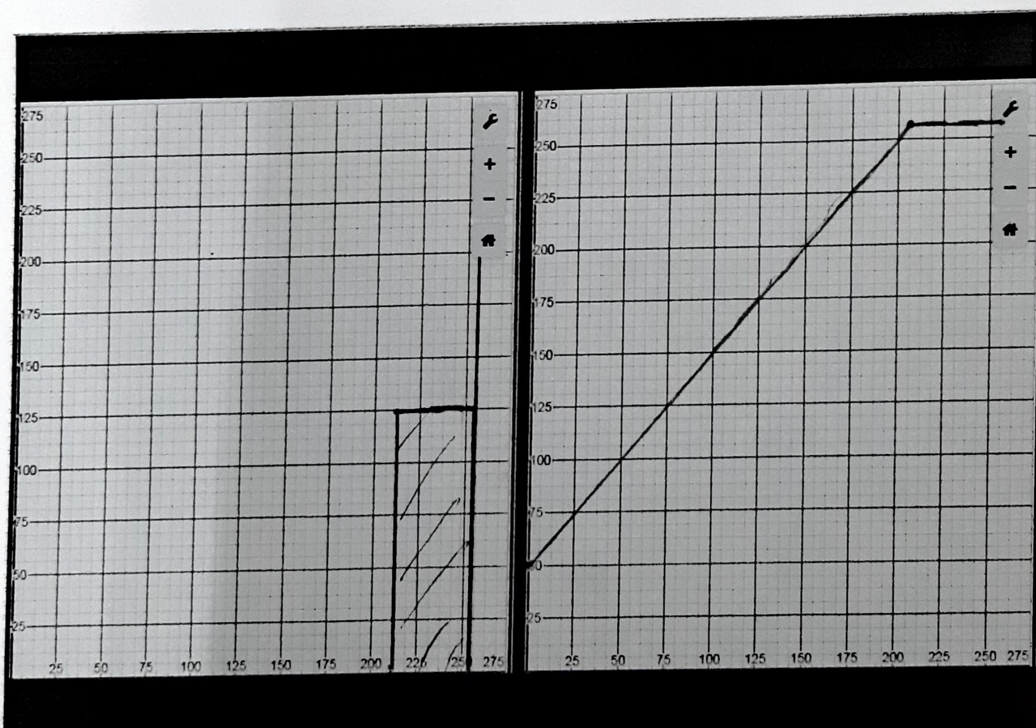
Histogram Transformation function

(a)



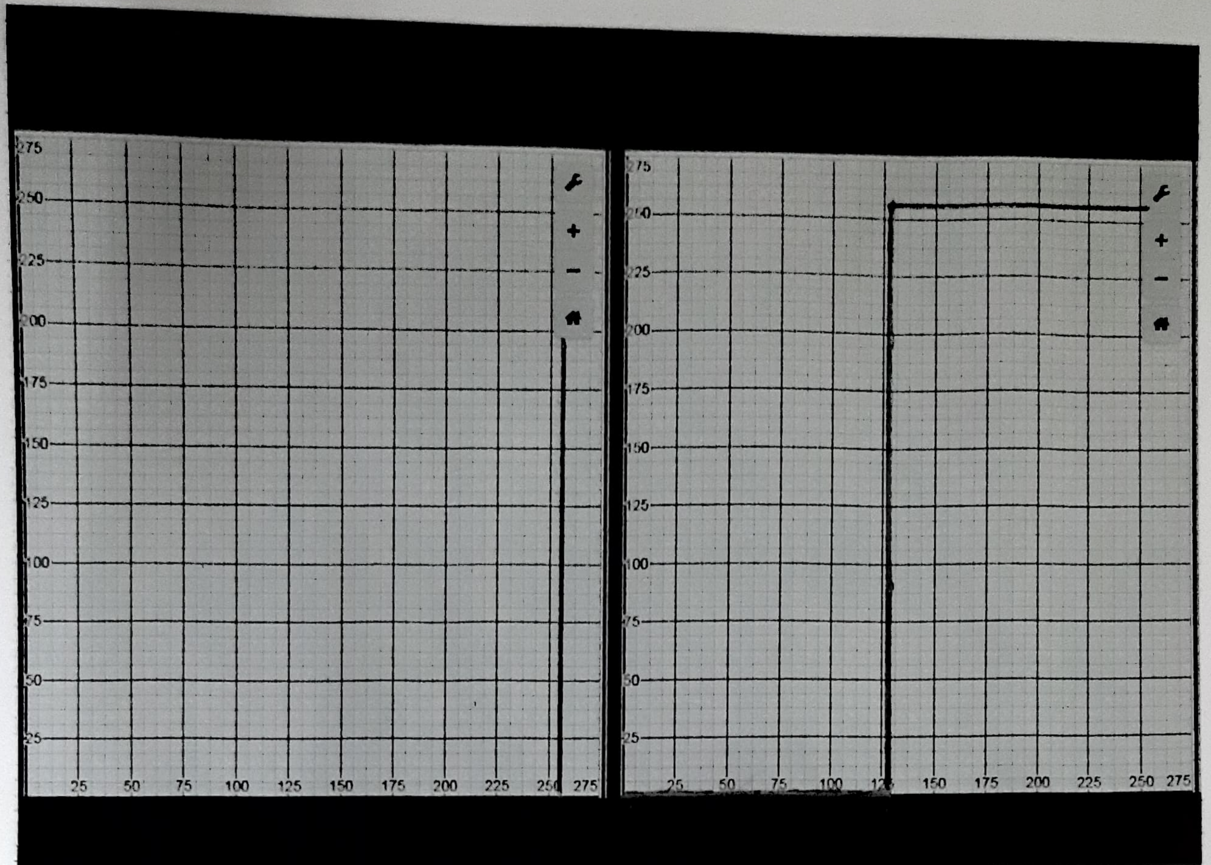
$S=(L-1-r)$

(b)



(b) This will lighten the image while maintaining a poor contrast as only a limited range of gray scale values are used. It will force all pixels which previously had a grayscale values between 206 and 220 to grayscale 255. This will make more pixels have value 255, reducing the contrast. The histogram ranges from 210 to 254 uniformly but 255 now has more values. The transformation is a line starting at point (0,50) ending at (205,255) and keeps a 255 value after 205.

(c)



(c) This will make the entire image white as the threshold is set below every gray scale value in the image. The new histogram will have values only at 255. The transformation function is a typical threshold.

IX. Suppose you want to send two episodes of one hour each of Big Bang Theory to your friend. In HD video the frame size is 1920 x 1080. The frames are sent at a rate of 30 fps. The display is 8-bit RGB. How many bits will you have to send? [4]

Total Time = 2 hrs = 7200 secs

Frame rate = 30 fps

Total Frames = 30 × 7200 = 216000 frames.

Frame size = 1920 × 1080.

8 bit RGB = 24 bit pixel

Total pixels in 1 frame = 2073600 pixels

Total bits in 1 frame = 49766400 bits

Total bits in 216000 frames = 216000 × 49766400 bits

$= 1.07495424 \times 10^{13}$ bits Ans